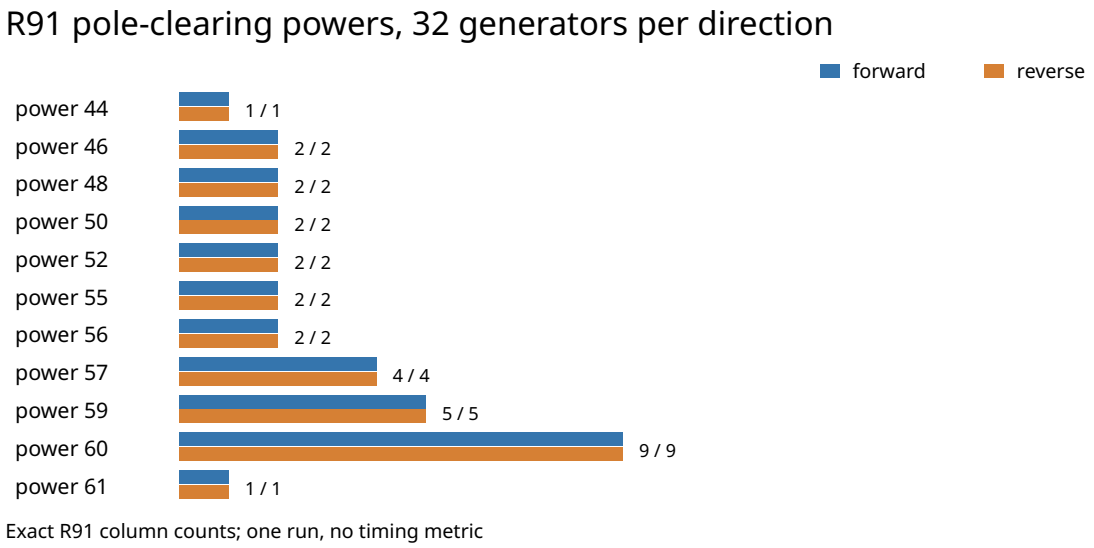
R91 bidirectional rank-32 RR transport

The single frozen R91 attempt checked all 32 normalized R50 generators in both rational diamond directions. The exact native gate passed 64 column products and 64 corrupted-coordinate controls. Independent replay from a fresh S3 readback checked both full products, 64 columns and 2,048 polynomial module coordinates in an ideal-free coefficient field.



The first pole-clearing powers range from 44 to 61 in each direction. The forward and reverse 32-value sequences coincide; one generator in each direction attains the bound 61. These are exact counts, not timings. Source revision `7ff259920`; seed 91; one native Singular/FLINT thread; 4 GiB virtual-memory cap; 1200-second timeout. All 18 raw members passed fresh-readback hashes. Earlier polynomial-ring replay probes failed because exported power vectors include rational coefficients; the corrected ideal-free F2(u) replay passes. The Lean module proves only the conditional integer pole inequalities. Proper boundary normalization and the modular-factor action remain the next constructions.